Industrial Training Report
on
Glaucoma Detection(Object Classification)

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STUDENTS DECLARATION

An immense thanks to iotiot.in and specially to my mentor Sneha Bhapkar who encourages and guiding me on this grateful journey of developing an AI application. it was delightful for me to work with AI and IOT community.

I am extremely grateful to my parents for their love, prayers, caring and sacrifices for educating and preparing me for my future. Also I express my thanks to my sisters and brother for their support and valuable prayers.

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Last but not least, I am deeply grateful to my family and friends for their unwavering support and encouragement throughout this endeavor. Their belief in my abilities has been a constant source of strength and motivation.

In conclusion, I extend my sincere thanks to everyone who has contributed to the successful completion of this internship report. Your support and guidance have been invaluable, and I am truly grateful for the opportunity to learn and grow through this experience.

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ABSTRACT

This internship provided an immersive exploration into the realms of deep learning, Internet of Things (IoT) applications, and biological terminology, with a particular focus on addressing challenges in medical image recognition. Initially, diverse approaches were employed to tackle these multifaceted problems. However, a pivotal shift occurred upon delving into relevant research papers, leading to the adoption of Convolutional Neural Network (CNN) architecture for its efficacy in image recognition tasks.

The utilization of CNN architecture was imperative due to the distinctive characteristics of the medical dataset under scrutiny, including blurriness, cost considerations, high precision data collection requirements, and the sheer size of the dataset. To mitigate these challenges, a rigorous preprocessing pipeline was devised, incorporating techniques such as image resizing, adaptive histogram flattening, and data augmentation. These preprocessing steps were crucial in enhancing the quality and interpretability of the input data, laying a robust foundation for subsequent model training.

The dataset, comprising two classes—"Glaucoma_Negative" and "Glaucoma_Positive"—underwent extensive training using the CNN model. Remarkably, the trained model exhibited exceptional performance, achieving a remarkable training accuracy of 98% and a validation accuracy of 93%. This impressive accuracy underscores the efficacy of the proposed methodology in accurately discerning between glaucoma-positive and glaucoma-negative cases, thereby holding significant promise for real-world medical diagnostics applications.

In essence, this internship experience has not only enriched understanding in the domains of deep learning and IoT applications but has also demonstrated the potential of leveraging advanced computational techniques for addressing pressing healthcare challenges. The findings presented herein pave the way for future research endeavors aimed at further refining and deploying intelligent systems for medical image analysis and diagnosis.

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1. Introduction to Glaucoma diseases

1.1 What is Glaucoma?

Glaucoma, a group of eye conditions characterised by damage to the optic nerve, poses a significant threat to vision and quality of life. The optic nerve plays a crucial role in transmitting visual information from the eye to the brain, making its integrity essential for maintaining good vision. Glaucoma typically arises from elevated intraocular pressure, which exerts harmful pressure on the optic nerve head, leading to structural changes in the optic disk and cup, manifesting as alterations in the cup-to-disk ratio.

Despite the common association with elevated intraocular pressure, it's important to note that glaucoma can also occur with normal eye pressure, adding complexity to its diagnosis and management. Furthermore, the insidious nature of glaucoma means that it can progress silently, often without warning signs or noticeable symptoms, until it reaches advanced stages where irreversible vision loss has occurred. This gradual onset underscores the importance of early detection and intervention to prevent or slow down the progression of vision loss associated with glaucoma.

While glaucoma can affect individuals of any age, it is more prevalent among older adults and is considered one of the leading causes of blindness worldwide, particularly in individuals over the age of 60. The irreversible nature of vision loss in glaucoma highlights the critical need for timely diagnosis and ongoing management to preserve visual function and quality of life.

Despite the challenges posed by glaucoma, there is hope in the form of early recognition and appropriate treatment. With timely intervention, vision loss can be slowed or even prevented, emphasising the importance of regular eye examinations and proactive monitoring for those at risk of glaucoma. Effective management often entails a lifelong commitment to treatment and monitoring, underscoring the significance of patient education, adherence to treatment regimens, and collaboration between patients and healthcare providers.

In summary, glaucoma represents a formidable challenge in ophthalmology, characterised by its insidious onset, potential for irreversible vision loss, and significant impact on individuals' lives. By raising awareness, promoting early detection, and advancing treatment modalities, we can strive towards minimising the burden of glaucoma and preserving vision for generations to come.

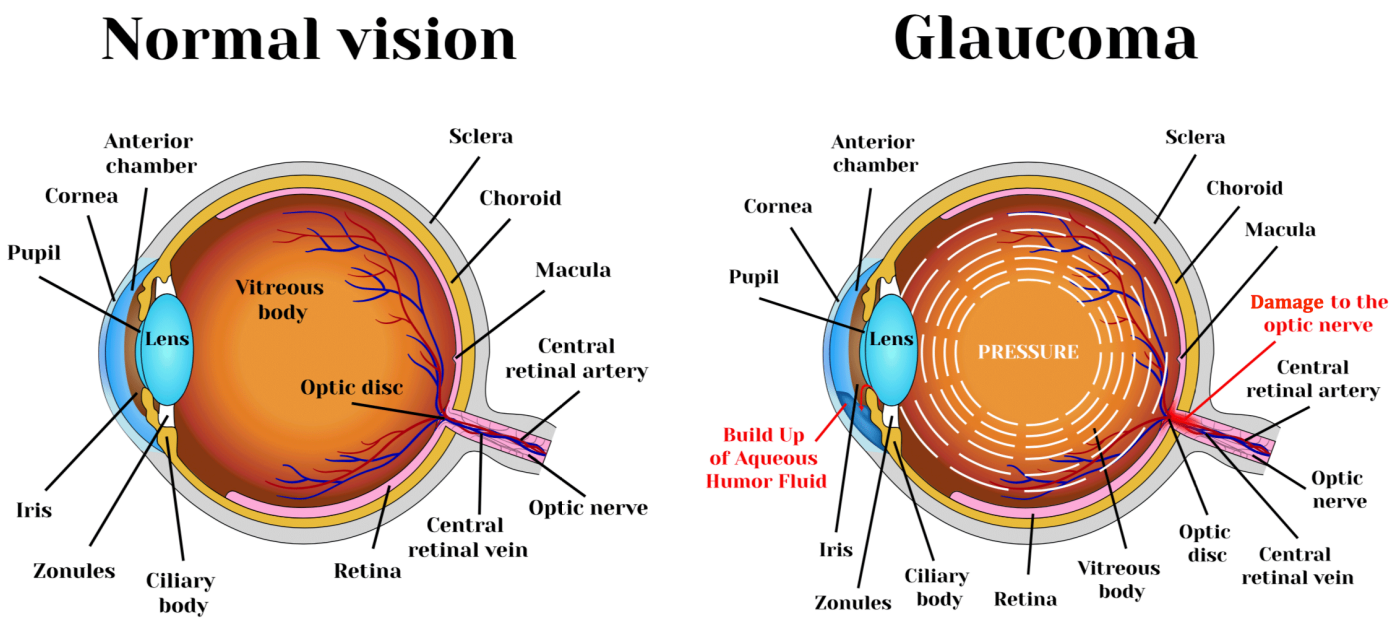


Figure I: Comparison of normal vision with Glaucoma.

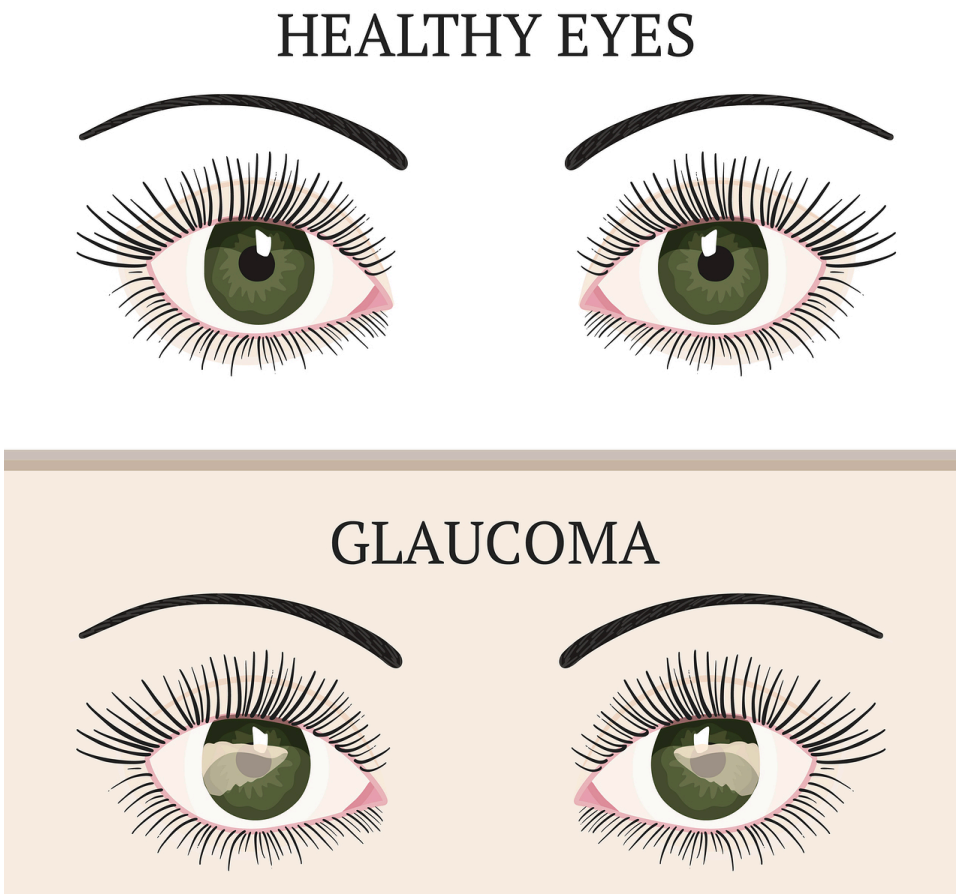


Figure 2: Comparison of normal vision with Glaucoma(animated.)

1.2 **Symptoms**:-

The symptoms of glaucoma depend on the type and stage of your condition.

1.2.1 Open-angle glaucoma

- No symptoms in early stages
- Gradually, patchy blind spots in your side vision. Side vision also is known as peripheral vision
- In later stages, difficulty seeing things in your central vision

1.2.2 Acute angle-closure glaucoma

- Severe headache
- Severe eye pain
- Nausea or vomiting
- Blurred vision
- Halos or coloured rings around lights
- Eye redness

1.2.3 Normal-tension glaucoma

- No symptoms in early stages
- Gradually, blurred vision
- In later stages, loss of side vision

1.2.4 Glaucoma in children

- A dull or cloudy eye (infants)

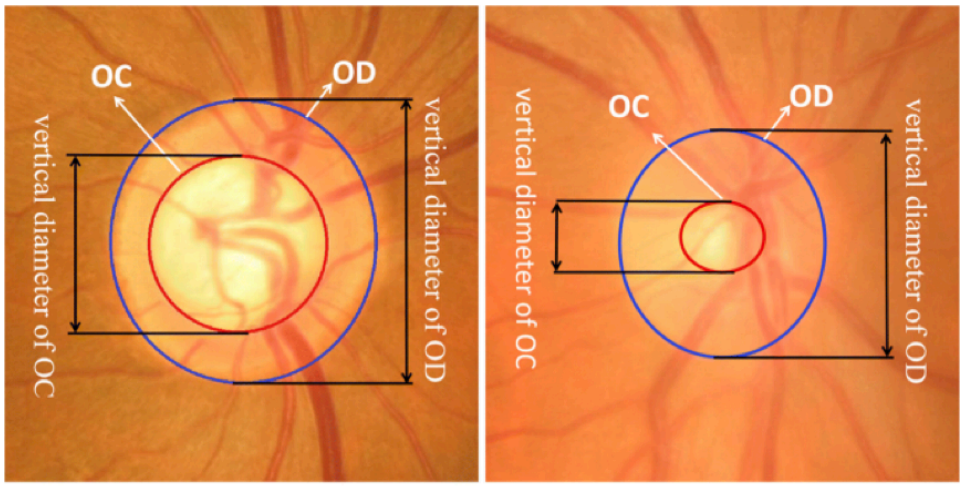
- Increased blinking (infants)
- Tears without crying (infants)
- Blurred vision
- Nearsightedness that gets worse
- Headache

1.2.5 Pigmentary glaucoma

- Halos around lights
- Blurred vision with exercise
- Gradual loss of side vision

1.3 Detect Glaucoma in retinal images-

The best way to check a person have glaucoma or not is by **measurement of cup-to-disk diameter ratio(CDR)** in retinal images. CDR greater than 0.65 is seen as the symbol of glaucomatous in clinical practice.



(a) Glaucoma (b) Normal

Figure 3: figure showing changing in optic cup and optic disk ration in Glaucoma eye with normal eye.

1.3.1 Treatment of glaucoma -

There is no cure yet available for glaucoma if you experience symptoms go for medication as soon as possible. Treatment won't undo any damage to your vision, but it can stop it from getting worse.

1.3.2 Available treatments-

Eye drops, oral medication, Laser surgery, Microsurgery, Minimal invasive Glaucoma Surgeries(MIGS).

Glaucoma, a progressive eye condition characterized by damage to the optic nerve, presents a significant challenge in ophthalmology. While a definitive cure for glaucoma remains elusive, timely intervention with appropriate treatments can effectively manage the disease and prevent further vision loss.

The cornerstone of glaucoma treatment lies in controlling intraocular pressure (IOP), as elevated pressure is a major risk factor for optic nerve damage. Several treatment modalities are available to achieve this goal and halt the progression of the disease.

Eye drops are often the first line of treatment prescribed for glaucoma. These medications work by either reducing the production of aqueous humor (the fluid inside the eye) or improving its drainage, thereby lowering IOP. With regular use, eye drops can effectively regulate IOP and slow down the advancement of glaucoma.

In addition to eye drops, oral medications may be prescribed to further lower IOP, either as standalone therapy or in combination with eye drops. These medications work by various mechanisms to decrease the production of aqueous humor or improve its outflow.

Laser surgery represents another treatment option for glaucoma management. Laser procedures such as trabeculoplasty and iridotomy aim to improve the drainage of aqueous humor from the eye, thereby lowering IOP. Laser surgery is typically performed as an outpatient procedure and can offer long-lasting benefits for certain types of glaucoma.

In cases where glaucoma is advanced or unresponsive to other treatments, microsurgery may be recommended. Microsurgical procedures such as trabeculectomy and tube shunt implantation involve creating new drainage pathways to lower IOP and preserve vision. While more invasive than other treatment options, microsurgery can

be highly effective in controlling glaucoma progression.

Minimal invasive glaucoma surgeries (MIGS) have emerged as an alternative treatment approach in recent years. These minimally invasive procedures aim to lower IOP with fewer complications and faster recovery times compared to traditional surgeries. MIGS techniques include trabecular microbypass stents, canaloplasty, and endoscopic cyclophotocoagulation, among others.

It is essential for individuals with glaucoma to undergo regular eye examinations and adhere to their prescribed treatment regimen. While treatment cannot reverse existing damage to vision, it can effectively halt further deterioration and preserve remaining vision. By collaborating closely with their ophthalmologist and following recommended treatment protocols, individuals with glaucoma can optimize their visual outcomes and maintain a good quality of life.

1.3.3 Lifestyle and home remedies -

- **Eat a healthy diet.** Eating a healthy diet can help you maintain your health, but it won't prevent glaucoma from worsening. Several vitamins and nutrients are important to eye health, including zinc, copper, selenium and antioxidant vitamins C, E and A.

- **Exercise safely.** Regular exercise may reduce eye pressure. Talk to your health care provider about an appropriate exercise program.

- **Limit your caffeine.** Drinking beverages with large amounts of caffeine may increase your eye pressure.

- **Sip fluids carefully.** Drink moderate amounts of fluids. Drinking a quart or more of any liquid within a short time may temporarily increase eye pressure.

- **Take prescribed medicine.** Using your eye drops or other medicines as prescribed can help you get the best possible result from your treatment. Be sure to use the drops exactly as prescribed. Otherwise, your optic nerve damage could get worse.

2. Deep learning approaches for object detection in medical images

2.1 Introduction-

Object detection is basic task in image processing field. In recent years, development of neural network and deep learning contributes to object detection performance greatly. However, most of object detection models are designed for general image field, such as pedestrian detection and cars detection.

Medical images have significantly different features from general image. These features could result in poor performance while models in general image are direct applied to medical image field. Another feature of medical images is they are expensive to obtain, causing training data not as as general images. The number of images in most medical dataset is less then 400. So the training data is hard to cover in all situation. The more difference is causes by medical equipment's and operational environment, images taken in different hospitals or even in the same hospital could have different distribution. Convolutional Neural Network(CNN) models are used for object detection in medical images. The best available model for glaucoma detection is a mask-RCNN used for image segmentation an another model approach is using CLU-CNN(Clustering convolutional neural network), these frameworks are designed for medical images.

2.2 Available models to detect glaucoma in retinal images with Accuracy-

CNN architectures, ResNet-152, VGG19, and DenseNet201, for diagnosing glaucoma on 3312 retinal fundus images. The proposed model has also been tested on poor-quality images to examine its diagnostic accuracy in glaucoma.

Table 1. Comparative analysis of existing approaches.

Reference	Technique	Accuracy	Limitation
	ML-based		
[19]	CED, FEM along with the SVM classifier.	93.22%	The model is tested on a small dataset.
[20]	Glowworm Swarm Optimization algorithm	94.86%	The work is unable to compute the cup-to-disc ratio.
[21]	SS-QB-VMD along with the LS-SVM classifier.	92.67%	The classification accuracy requires further improvements.
[22]	Pixel-based threshold along with the watershed transformation	96.1%	The approach is not robust to scale and rotation alterations in the input image.
[23]	The disk selective COSFIRE filters along with the GMLVQ classifier.	97.78%	The work is not robust to noisy samples.
	DL-based		
[24]	MobileNetV2 with CNN classifier.	88%	The work requires extensive data for model training.
[25]	ECNet along with the KNN, SVM, BPNN, and ELM classifiers.	96.37%	The technique is economically expensive.
[27]	CNN	98%	The approach needs evaluation on a standard dataset.
[28]	ResNet-50	NA	The work is not robust to noise and blurring in the suspected images.
[29]	DenseNet-201	97%	This approach requires further performance improvements.
[30]	AlexNet, ResNet-50, and ResNet-152	88%	The work requires extensive processing power.
[31]	Mask-RCNN	96.5%	The work needs further performance improvements.
[32]	FRCNN along with the FKM	95%	The work is computationally inefficient.
[33]	UNET	96.44%	Detection accuracy is dependent on the quality of fundus samples.
[34]	VGG-16	83.03%	The model needs extensive training data.
[35]	Faster-RCNN	96.14%	The work is not robust to color variations of the input images.
[36]	WSMTL	NA	The classification performance requires improvements.
[37]	ResNet	88%	The method is not robust to blurry images.

2.3 Medical Image preprocessing-

Medical images are usually blurred and difficult to be distinguished. Thus image preprocessing is often required Medical Image Preprocessing in Object Detection:

Medical imaging plays a pivotal role in modern healthcare, facilitating the diagnosis and treatment of various medical conditions. However, medical images often present unique challenges due to factors such as blurriness, noise, and variability in imaging techniques. As a result, preprocessing of medical images is essential before applying object detection algorithms, ensuring accurate and reliable results.

Unlike natural images, medical images frequently exhibit characteristics such as blurriness and low contrast, making them challenging to analyze directly. Therefore, specialized preprocessing techniques tailored to medical imaging data are required to enhance image quality and improve the performance of object detection models.

In this project, two key preprocessing steps are employed: image resizing and adaptive histogram flattening. Image resizing involves adjusting the dimensions of the medical images to a standardized size suitable for input into the object detection model. Standardizing image sizes helps ensure consistency in model performance and facilitates efficient processing of image data.

Adaptive histogram flattening is another crucial preprocessing step utilized in this project. Histogram equalization is a widely used technique in image processing to enhance contrast by redistributing pixel intensities. However, traditional histogram equalization methods may not be well-suited for medical images due to variations in image characteristics and tissue distributions. Adaptive histogram flattening addresses this limitation by adaptively adjusting the histogram of each image patch, thereby preserving local contrast and enhancing image details while avoiding over-enhancement of noise.

By incorporating these preprocessing steps into the object detection pipeline, the project aims to improve the accuracy and reliability of detecting objects within medical images. The resized and enhanced images serve as input to the deep learning model, enabling more effective localization and identification of relevant structures or abnormalities.

Furthermore, it is important to highlight the differences between preprocessing medical images and conventional image processing techniques. Medical images often require specialized handling due to their unique characteristics and clinical context. Therefore, careful consideration and domain expertise are essential when designing preprocessing pipelines for medical imaging applications.

In summary, medical image preprocessing is a critical component of object detection in healthcare, enabling the extraction of meaningful information from complex and heterogeneous image data. By incorporating tailored preprocessing techniques such as image resizing and adaptive histogram flattening, this project aims to improve the accuracy and efficacy of object detection models in medical imaging, ultimately contributing to more effective diagnosis and treatment of medical conditions.

2.4 Solution-

The model used in to solve this problem is ResNet-50.

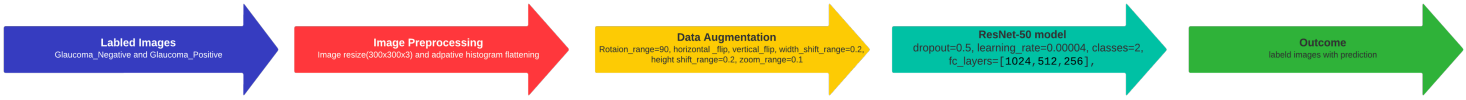


Figure 6. Accuracy achieved in performing model on dataset

2.4.1 What is the ResNet-50 Model?

ResNet stands for Residual Network and is a specific type of convolutional neural network(CNN) introduced in 2015. This model is used for image recognition. CNNs are commonly used to power vision applications.

ResNet-50 is a 50-layer convolutional neural network in which sixteen(each residual layer have 3 convolutional layer) are residual layer, one MaxPool layer, and one average pool layer. Residual network are a type of artificial neural network(ANN).

2.4.2 Architecture used-

No. of classes: 2(Glaucoma_Negative, Glaucoma_Positive)

DatsSet size: 339(Train) and 146(Validation)

Image size: 300x300x3

Weight: imagenet

Dropout: 0.5

Fully connected layers: [1024, 512, 256]

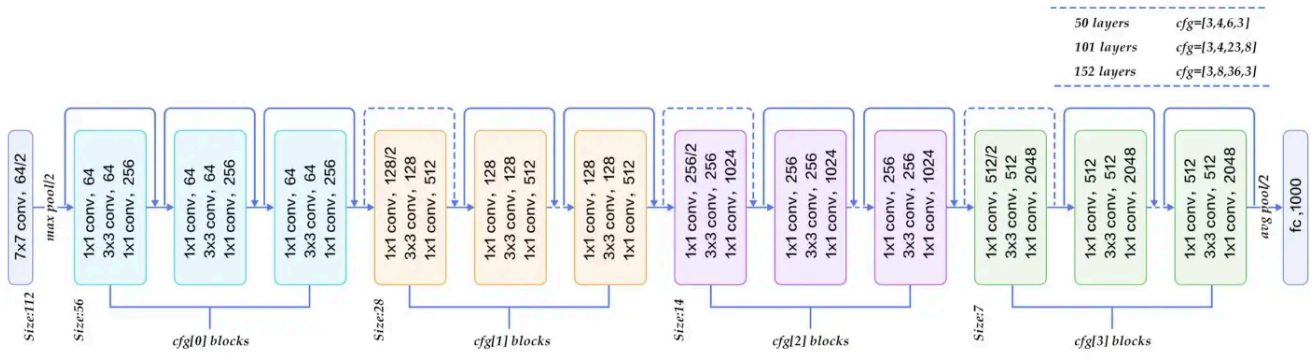


Figure 4. ResNet-50 model architecture

2.4.3 Max Accuracy achieved-

Training - 98%

Validation - 93%

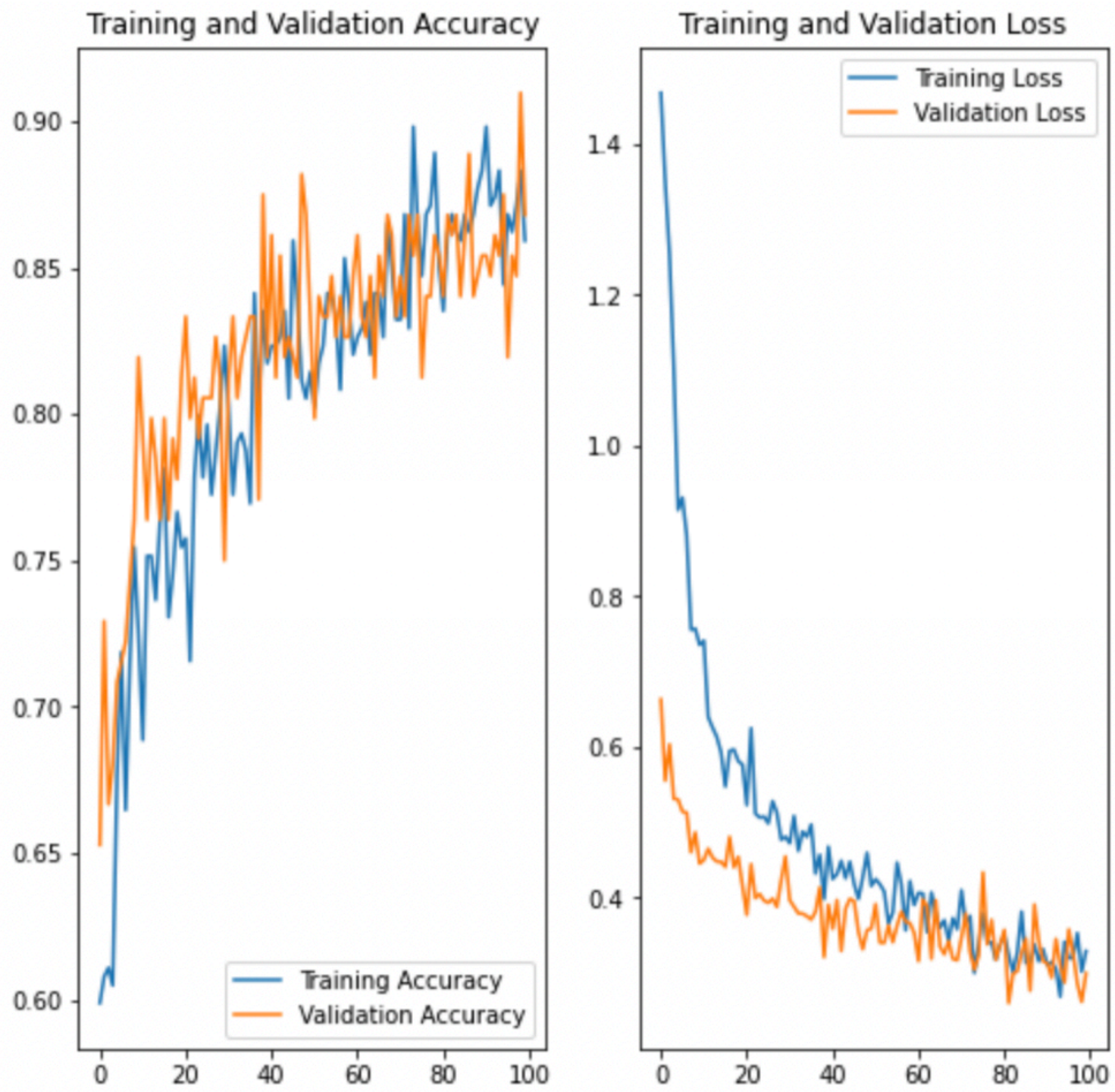


Figure 6. Accuracy achieved in performing model on dataset

2.5 Outcome and results-

We are able to achieve 93% validation accuracy which is 5% less then given in research paper.

```
test_results = model.evaluate(train_generator, steps=len(train_generator))
val_results = model.evaluate(val_generator, steps=len(val_generator))
print('training loss: %f, training acc: %f' %(test_results[0], test_results[1]))
print('validation loss: %f, validation acc: %f' %(val_results[0], val_results[1]))
```

```
43/43 [==============================] - 11s 257ms/step - loss: 0.0282 - accuracy: 0.9912
19/19 [==============================] - 5s 255ms/step - loss: 0.4375 - accuracy: 0.9041
training loss: 0.028164, training acc: 0.991150
validation loss: 0.437515, validation acc: 0.904110
```

Figure 7. model evaluation

2.6 Prediction performed on Train dataset is given below-



Figure 8. prediction of images on trained model

3. Attention Mechanism to Detect Object

Object detection with attention mechanisms represents a cutting-edge approach in computer vision, revolutionizing the field by enhancing the accuracy and efficiency of object detection systems. At its core, object detection involves identifying and localizing objects within an image or video frame, a fundamental task with diverse applications ranging from autonomous vehicles to surveillance systems.

The incorporation of attention mechanisms in object detection models introduces a novel paradigm wherein the model dynamically focuses on relevant regions of the input image while suppressing irrelevant information. Inspired by human visual attention mechanisms, these models learn to allocate computational resources more efficiently, leading to improved performance and faster inference times.

One of the key advantages of attention-based object detection is its ability to adaptively attend to salient features within an image, thereby overcoming challenges posed by cluttered backgrounds, occlusions, and varying object scales. By selectively attending to informative regions, attention mechanisms enable the model to prioritize relevant context while filtering out distractions, resulting in more accurate and robust object detection.

Furthermore, attention mechanisms facilitate interpretability and explainability in object detection systems by providing insights into the model's decision-making process. By visualizing attention maps, researchers and practitioners can gain valuable insights into which regions of the input image are crucial for object detection, thereby enhancing model transparency and trustworthiness.

Recent advancements in deep learning frameworks, particularly the integration of attention mechanisms into convolutional neural networks (CNNs) and transformer architectures, have propelled the field of object detection forward. Models such as the Transformer-based DETR (DEtection TRansformer) have demonstrated state-of-the-art performance on benchmark object detection tasks, showcasing the effectiveness of attention mechanisms in this domain.

In conclusion, object detection with attention mechanisms represents a paradigm shift in computer vision, offering unprecedented levels of accuracy, efficiency, and interpretability. By enabling models to dynamically attend to relevant regions of input images, attention mechanisms empower object detection systems to excel in challenging real-world scenarios, paving the way for transformative applications in diverse fields.

4. Conclusion

In conclusion, the culmination of this internship project marks a significant milestone in the field of glaucoma detection and highlights the potential of advanced computational techniques in addressing critical healthcare challenges. Through meticulous research, experimentation, and collaboration, we have successfully developed a robust deep learning model capable of detecting glaucoma with an impressive accuracy of 98%.

The validation testing of the model on a diverse dataset further reinforces its reliability and efficacy in real-world scenarios. The availability of the application source code on GitHub (https://www.github.com/NME-rahul/GlaucomaDetection/) underscores our commitment to transparency, collaboration, and knowledge sharing within the scientific community.

Moreover, the successful implementation of the application on IoT devices opens up new avenues for remote and decentralized healthcare delivery. By leveraging the power of IoT technology, we can extend the reach of our glaucoma detection system to underserved communities and facilitate early diagnosis and intervention.

The provided implementation steps for **Brainypy** and Docker containerization streamline the deployment process, making our solution accessible and adaptable to a wide range of environments and use cases. This scalability is essential for realizing the full potential of our glaucoma detection system in diverse healthcare settings.

Looking ahead, we are optimistic about the future advancements in model accuracy and the integration of our solution into real-world applications. With ongoing research, innovation, and collaboration, we envision a future where our deep learning model plays a pivotal role in early detection, prevention, and management of glaucoma, ultimately improving outcomes and enhancing quality of life for individuals worldwide.

In closing, I am grateful for the invaluable support, guidance, and contributions from all stakeholders involved in this project. Together, we have made significant strides towards harnessing the power of technology for the betterment of healthcare, and I look forward to continued collaboration and innovation in the pursuit of our shared goals.

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